

Seminar Thesis on *Generating Computational Cognitive Models using Large Language Models*

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Motivation

Computational cognitive models provide a formal framework for understanding human behaviour. In practice, researchers hand-craft these models, fit them to behavioural data, and iteratively refine them. This process is slow, skill-intensive, and systematically biased due to a person's past exposure to the cognitive literature. Translating a psychological theory into a working algorithm requires lengthy literature reviews and repeated coding cycles. Researchers must simultaneously act as domain experts in cognitive science, theoreticians for data modeling, and software engineers for implementation. Perhaps most importantly, researchers tend to explore only the model classes they already know and favour, potentially missing alternative explanations for the data. This resembles the *streetlight effect*: investigation is pulled toward the places where existing tools and theories make searching easiest. The aforementioned limitations narrow the space of models considered, motivating an automated approach that is less tied to any single school of

thought in theoretical cognitive science.

Three specific LLM capabilities make them well-suited for proposing cognitive models for experimental data.¹ First, LLMs can process task descriptions and behavioural data written in natural language, which makes them highly adaptable across different experimental paradigms without the need to provide a custom interface. Second, using broad representational and reasoning abilities, they can identify structure in complex behavioural data and generate plausible psychological hypotheses by reasoning over task structure and behavioural patterns. Third, LLMs can translate those hypotheses directly into executable Python functions. These three abilities have already been utilized in automated statistical modelling through probabilistic program generation, and [Rmus et al. \(2025\)](#) apply the same logic to cognitive science.

Proposed Pipeline - GeCCo

As illustrated in Figure 1, each pipeline run consists of 10 sampling iterations, repeated across 5 independent runs per aforementioned cognitive domain. The process begins by providing the base LLM with a fixed prompt containing a natural language task description, a subset of behavioral data, strict instructions, a domain-specific code template, and starting from the second iteration onwards, iterative feedback. Using this information, the LLM generates three unique cognitive models, ensuring that each features a non-overlapping set of parameters. Each full run took at most eight hours on four Nvidia A100 GPUs (40GB each), which is modest compared to the days typically required by search-based or evolutionary alternatives.

Once generated, these models are parsed from the response, converted into executable Python functions, and have their specific parameter names extracted. They are then fitted to a held-out validation dataset that was deliberately excluded from the initial prompt. To prevent the models from getting stuck in local minima, this fitting process is initialized from 20 random starting points.

After the fitting stage, the models are evaluated using the Bayesian Information Criterion (BIC) to identify the single best-performing candidate among the three. Finally, the code for this winning model, along with a list of previously used parameter names, is gathered to act as feedback. This feedback is then injected into the prompt for the next iteration, effectively guiding the LLM’s future proposals and preventing it from duplicating its past ideas.

Once all 10 sampling iterations are complete, the pipeline identifies the best model across the entire run. This winning model is then evaluated through two final verification steps:

Baseline Model Comparison: The LLM-generated model is fitted to the behavioral data, and its BIC score is directly compared to an established baseline cognitive model. This proves whether the LLM’s novel theory mathematically outperforms human-crafted theories.

Posterior Predictive Checks: To further verify validity, the LLM-generated model is

¹The models used were: Llama-3.1-Instruct-70B, DeepSeek-R1-Distill-Llama-3.1-70B, and Qwen2.5-72B-Instruct.

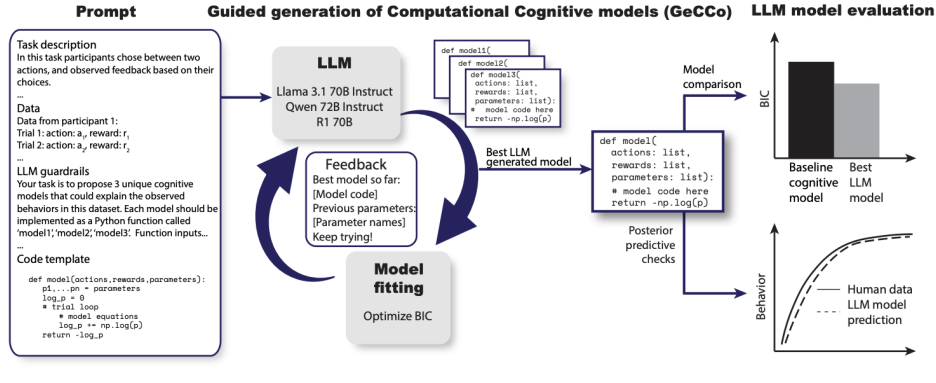


Figure 1: Graphical illustration of the pipeline proposed by [Rmus et al. \(2025\)](#).

run as a simulation. Its predicted behavioral outputs are plotted against the actual, ground-truth human data to ensure the model’s behavior mimics real human patterns.

Discussion

The GeCCo pipeline successfully demonstrates that open-source LLMs can act as automated research assistants, generating and refining psychological models using standard Python code. Across four distinct areas of human cognition, these AI-generated models were broadly competitive with the best hand-crafted models built by human experts: they achieved lower BIC in all four domains, with the improvement confirmed by a significance test in decision making and working memory, while learning and planning showed smaller or non-significant BIC differences alongside favourable Exceedance Probabilities. Crucially, the AI did not merely reproduce known ideas; it frequently proposed simpler or alternative theoretical explanations. For example, the LLM discovered a single discount factor for decision-making heuristics, cleanly separated experienced from counterfactual outcomes in learning tasks, and suggested that cognitive load directly modulates the reinforcement learning rate in working memory tasks.

Because human cognitive modeling is often inherently limited by the theoretical assumptions researchers start with, they may overlook simpler explanations. GeCCo partially mitigates this streetlight effect by acting as a hypothesis-generation tool that expands the set of possible models. However, the final interpretation still relies on human scientific judgment. The control experiments provide evidence that this success is driven by the AI’s ability to learn from its mistakes through iterative feedback, rather than from data contamination. Although it is worth noting that several of these analyses, the feature and ablation regressions in particular, rest on a modest number of independent pipeline runs per domain. This limits how strongly the relative importance of each factor can be estimated. Furthermore, by evaluating the generated models on strictly held-out data that the LLM never sees during the prompting phase, the pipeline reduces the risk of overfitting. The AI’s resulting code remained highly interpretable to humans while capturing comparable predictable human behaviour like CENTAUR in most domains.

A major benefit of GeCCo is its high accessibility; the system works out of the box using standard, open-source AI models. It utilizes a hybrid approach where the AI acts as

an intelligent proposal engine to brainstorm the mathematical structures, but the actual testing of how well those models fit the data is done offline using standard rigorous statistical tools. Thus, the final model selection is determined by BIC on held-out data, while the proposal process remains a hybrid search procedure guided by iterative feedback from previous rounds.

Despite these results, the authors outline several limitations that chart the course for future research. One limitation is template bias; the AI is currently provided with a basic, human-made code template to guide its syntax, which might unintentionally restrict its creative exploration to specific model classes. Although, as noted above, the authors' own follow-up check using an LLM-generated template found comparable performance, which somewhat mitigates this concern. The system's current feedback mechanism relies purely on predictive performance metrics like the BIC. However, a model with a lower BIC is not automatically the best scientific explanation, and human researchers must still verify if the mechanisms are psychologically meaningful and biologically plausible; the authors suggest that future iterations could incorporate a secondary AI to provide qualitative feedback critiquing theoretical logic or biological plausibility.

Finally, the study is limited in scope because it evaluated few specific laboratory tasks. Future research must extend this pipeline to more complex areas. It is also vital to test GeCCo's capacity for cross-task generalization to ensure that an AI-generated cognitive theory can robustly explain behavior across related tasks rather than just fitting a single dataset. Ultimately, these findings suggest that LLMs have the potential to democratize the highly technical field of computational modeling and significantly accelerate the pace of scientific discovery.

Christian's Part

Comparison with Luo et al

In BrainBench, the study shows the LLM two versions of a scientific abstract, one real and one with altered results, and examines which one it finds less surprising. This is measured through perplexity; a lower perplexity means the text reads as more expected, and this is treated as the LLM's vote for which version is correct.

But the approach rests on an assumption that is not easily verifiable: that a passage feeling less surprising is the same thing as a passage being scientifically correct. The altered abstracts were created by editing the real ones, either by hand or by GPT-4 following a fixed set of instructions, and editing tends to leave traces. Instead of spotting bad science, there is a risk that the model is just noticing that one passage was edited and the other was not. The authors do run checks meant to rule this out, showing for instance that performance drops once context is stripped away, which suggests the model is drawing on more than just the altered sentence. That is reassuring, but it does not fully close off the edit-detection worry.

The headline result that LLMs outperform human experts also needs a caveat. Every LLM was tested on the full set of 200 items, but each human participant only saw nine or ten of the items. So the human score is not one expert's performance on the entire test, but rather an average built from many experts, each exposed to a different set of items.

The two papers use LLMs in ways that are not really comparable. GeCCo claims that an

LLM can come up with a workable scientific theory on its own. BrainBench, by contrast, hypothesizes that an LLM’s sense of what sounds right beats human intuition.

Reflection Use of AI

Alongside reading the paper itself, I used AI a fair amount throughout the process, though always as a tool to help me think. In the early stages I inserted excerpts from the GeCCo paper and asked for summaries and simpler explanations of the different experiments along with the control experiments. That helped me actually get comfortable with things like the Bayesian Information Criterion, the iterative feedback loops, and the prompt ablations.

Once I had a decent understanding on the paper itself, I used AI again to help me place it into a broader context. I asked it to compare GeCCo’s approach to AutoResearch, to BrainBench, and to the recursive self-improvement work coming out of Anthropic, Sakana AI, and Google DeepMind. That was useful for the discussion section, since it let us show that GeCCo is not just an isolated concept.

Later on I leaned on AI for structure than content. I asked for outlines, one for the 20-minute presentation, and one for the written report, and those outlines basically became the skeleton we built everything else around. Once we had that skeleton, I used AI to help turn my notes into proper academic language. Afterwards we went through the text again to make it our own language/style.

Reflection on the Reading

I started working on this thesis in mid-May, right after I got my paper assignment. I messaged my partner on Discord quickly and we agreed on a plan: read the paper on our own first, ask an LLM whenever we hit something confusing, and then meet up again a week later to actually talk it through. In the meantime, I started thinking about how I would structure the presentation.

When I sat down and read the paper properly, I tried to really understand what the paper was actually contributing. I also looked through the authors’ GitHub to see how they actually things. After that I wrote a first draft of the report and presentation.

the paper kept reminding me of Karpathy’s Autoresearch repo. That was not just a random thought either; it ended up shaping my view on the paper. Not long after that, I started seeing the GeCCo pipeline as connected to the whole recursive self-improvement debate that Anthropic and others were talking about. Suddenly the paper did not feel like some isolated entity, it felt like it was part of a bigger conversation happening currently.

By early June I finished the report with my partner, and along the way I finally understood how to interpret the different information criteria. Then in July, I read the paper again. This time it stood out to me how mathematically rigorous cognitive models actually are, e.g. Rescorla–Wagner model and the Drift Diffusion Model. Further I realized how important posterior predictive checks and exceedance probability are for evaluation purposes

YingZhu's Part

Comparison with Card et al

The two papers assign AI fundamentally different scientific roles. GeCCo uses an LLM as a *hypothesis generator*: it proposes executable cognitive models, while conventional optimization and BIC select among them. In Card et al. (2025), a transformer-based speech decoder and an RNN-based cursor decoder instead act as *assistive prediction systems*, mapping neural activity directly to words and computer movements. GeCCo therefore prioritises interpretable theory generation, whereas Card et al. prioritise predictive reliability and practical utility.

The evidence in Card et al. is strong for a narrow claim. One participant with ALS used the intracortical BCI at home for more than 3,800 hours over 19 months and produced 183,060 sentences. This unusually long observation period demonstrates sustained usefulness for this participant. However, it does not establish general clinical viability. The headline 99.2% word accuracy came from prompted copy tasks; during spontaneous use, accuracy had no independent ground truth, 92.3% of utterances were self-rated as at least mostly correct, and only 53.3% were rated completely correct without correction. Performance also reflected the language model, background recalibration, interface updates, and gaze-based correction, not only neural decoding. Moreover, the study involved one participant, percutaneous connections, and daily setup by trained care partners. Its ecological depth is therefore a major strength, but population-level generalisability remains unknown.

GeCCo has the opposite evidence profile: it covers four cognitive domains and produces readable models, but lower held-out BIC shows predictive usefulness rather than psychological truth. Card et al. demonstrate direct real-world value, but use less interpretable models and provide no replication across users. More convincing follow-up evidence would therefore differ: GeCCo needs preregistered cross-task tests of whether its mechanisms generalise, whereas the BCI needs a multi-participant study with objective measures of spontaneous-speech error, comparison with existing assistive devices, and systematic assessment of fatigue, setup burden, adverse events, and quality of life.

Reflection Use of AI

Throughout the course, I used generative AI to clarify technical terminology, understand statistical measures and model architectures, translate or rephrase selected passages, organize preliminary notes, and improve the language of the slides and report. AI also helped me identify possible comparison points and limitations. However, I did not treat its answers as academic evidence. Numerical results and methodological details were checked against the original papers, and AI-generated suggestions were revised or rejected when they were not supported by the text. The final evaluation of the papers' claims and evidence was completed by me and my group member.

Reflection on the Reading

At the beginning of the semester, our reading focused mainly on reconstructing what each model did. In the GeCCo paper, this meant understanding the iterative prompting loop, the separation of training, validation, and test data, and the meaning of BIC and

exceedance probability. Preparing the presentation changed our reading strategy: instead of treating every reported improvement as equivalent evidence, we traced each headline claim to the experiment that was supposed to support it. This revealed, for example, that significant BIC improvements were strongest in decision making and working memory, while the evidence in learning and planning was weaker. Reading the BCI paper reinforced the same lesson from the opposite direction. Its 19-month case study provides unusually rich evidence for sustained use by one participant, but the impressive amount of data does not increase the number of independent participants. We therefore learned to separate longitudinal depth from population-level generalisability, and benchmark accuracy from everyday usefulness.

References

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- Rmus, M., Mathony, M., Jagadish, A. K., Ludwig, T., and Schulz, E. (2025). Generating computational cognitive models using large language models. *arXiv preprint arXiv:2502.00879*.

Learning Journal and LLM logs - YingZhu Chen

April 23. This was my first class. I learned about the structure, content, and expectations of the course, as well as how the seminar would be organized. At this stage, I was still trying to understand what kind of work was expected from us. I also began thinking about research directions that might be relevant to my future bachelor's thesis.

April 30. I learned how to read academic papers in a more structured way. Instead of reading every section with the same level of detail, I learned to first identify the research question, main claim, method, evidence, and conclusion. I also learned how to select papers that are relevant to a specific research field. This helped me narrow down my interests and find papers that could support my future research.

May 4. I selected several papers related to my research interests. At first, I found them difficult to understand because they contained many unfamiliar technical terms and concepts. I searched for background explanations and used ChatGPT to clarify difficult terminology and selected passages. This also helped me learn relevant academic English vocabulary. However, I noticed that AI explanations could sometimes be too general, so I began checking them against the original papers.

May 11. After two seminar sessions, I had a clearer understanding of how to read a scientific paper and prepare an effective presentation. I continued reading Paper 15, which was the paper most closely related to my interests. I used ChatGPT to help identify the paper's structure and create an initial outline. This made the general pipeline easier to understand, but I still had difficulty interpreting the figures, statistical measures, and differences between the four experiments. In particular, I did not yet fully understand what a lower BIC score could and could not demonstrate.

May 16. After being assigned to a group, I discussed the next steps with my group

member. We decided to read the paper independently again and use an LLM when we encountered unfamiliar topics, especially concepts related to cognitive modelling, reinforcement learning, and neuroscience. I was responsible for preparing the first part of the presentation slides. We also arranged a Discord meeting for May 22. Before the meeting, each of us prepared a possible presentation structure so that we could compare our ideas and agree on a shared outline.

June 7. We completed an initial version of our report notes and held another group discussion. We identified the parts of the paper that we still did not fully understand and prepared questions for the professor. One problem was that our draft mainly described the GeCCo pipeline and the four experiments without clearly evaluating the authors' claims. We therefore began reorganizing the material around three questions: what the authors claim, what evidence they provide, and whether that evidence is sufficient.

June 16. We completed the first full version of the presentation slides. At this point, we noticed that the presentation contained too many methodological details and did not clearly distinguish between predictive performance and scientific explanation. For example, a generated model with a lower BIC may fit the data better, but this does not prove that its psychological mechanism is correct. We revised the slides to make this distinction clearer and selected the decision-making and working-memory experiments as the strongest examples. We kept the planning experiment as a contrasting case because its BIC improvement was not statistically significant.

June 19. We updated the presentation script and discussed it together. The main problems were weak transitions between sections, repeated explanations, and an unclear connection between the experimental results and our criticism. We shortened several descriptive passages and added sentences explaining what each experiment actually supported. We also clarified the role of AI in GeCCo: the LLM proposes candidate models, while conventional statistical tools fit and evaluate them. This distinction helped us avoid presenting the LLM as if it independently completed the entire scientific process.

July 6. We held our final discussion about the presentation and rehearsed the complete script. We checked the timing, division of speaking parts, and transitions between speakers. During the rehearsal, we found that some claims still sounded too strong, particularly the statement that GeCCo "outperformed" established models in all four domains. We revised this to explain that the generated models had lower mean BIC scores in all four domains, but the statistical evidence was stronger in decision making and working memory than in learning and planning. We also prepared possible answers to questions about overfitting, data contamination, template bias, and psychological plausibility.

July 9. We gave our presentation. Preparing and presenting the paper changed the way I evaluate scientific work. At the beginning of the semester, I mainly focused on understanding the model architecture and reported results. By the end, I was more interested in whether the experiments actually supported the authors' broader claims. I learned that strong predictive performance can support a model as a useful candidate, but additional experiments are needed before accepting it as a scientific explanation.

July 12. I reviewed Paper 15 again and compared it with my second paper, which examined the long-term independent use of an intracortical BCI. This comparison helped me understand that AI can have very different scientific roles. In GeCCo, AI generates candidate cognitive theories; in the BCI paper, AI decodes neural activity into speech and

cursor movements. I also learned to distinguish between different types of evidence. The BCI paper provides unusually detailed evidence from 19 months of use, but it includes only one participant, so it cannot establish general clinical effectiveness.

Learning Journal and LLM logs - Christian Block

Learning Journal

30.04.26. I submitted my preferences for the papers.

15.05.26. I received my paper to work on the day before yesterday. I got in contact with my partner for the thesis on Discord yesterday. We are now deciding on a plan how to approach the thesis to have something to discuss during the first office hour in June.

16.05.26. First meeting via Discord to discuss logistics and our goals as a team for the seminar thesis. We decided to read the paper on our own again and receive help from an LLM on unfamiliar topics from Neuroscience, and meet on Friday, 22nd of May, to discuss the paper together on Discord. In the meantime everybody conceives a structure for the presentation which we are going to discuss.

22.05.26. I am reading the paper seriously today to grasp the contribution and get an idea how to present it to my peers. Furthermore, I am looking into the author's GitHub repository.

23.05.26. I created the first draft for the report and presentation. We now have to refine it based on our team's goals and personal preferences.

31.05.26. The more I think about our assigned primary paper the more it reminds me of Karpathy's Autoresearch repository on GitHub.

06.06.26. Regarding the GeCCo pipeline, it seems highly related to recursive self-improvement, which Anthropic and many others are currently talking about.

07.06.26. Finished the report more or less; I learned the interpretation of many Information Criteria.

07.07.26. I read the paper once again; I was intrigued how mathematically rigorous the models in cognitive science are, e.g. the Rescorla-Wagner model, the Drift Diffusion Model, etc.

08.07.26. I finally got a good overview of many performance metrics: posterior predictive check and exceedance probability.

09.07.26. I learned a lot from the other students' presentations, especially how researchers in neuroscience handle high-dimensional data with encoders.

Stage	Prompt Intent / Interaction	Usage of AI Output
1. Comprehension & Analysis	Provided abstracts and excerpts from the GeCCo paper. Requested summaries, layman explanations of specific experiments (Decision Making, Learning, Planning, Memory), and elaborations on the control experiments.	Output was used to solidify personal understanding of complex methodologies, including the Bayesian Information Criterion (BIC), iterative feedback loops, and prompt ablations.
2. Contextualization	Asked to compare GeCCo’s methodology with AutoResearch, BrainBench (Luo et al.), and recursive self-improvement models from Anthropic, Sakana AI, and Google DeepMind.	Output was used to situate the GeCCo framework within the broader, current landscape of automated scientific discovery for the discussion section.
3. Outlining	Requested structural outlines for a 20-minute presentation suitable for a general audience, as well as a comprehensive structure for the written report.	Output served as the foundational blueprint for organizing the final seminar thesis and accompanying slide deck.
4. Drafting & Formatting	Requested the translation of synthesized summaries into formal academic prose without bullet points, specifically formatted as compiled L ^A T _E X code.	The generated L ^A T _E X code was critically reviewed, edited for accuracy, and integrated into the final Overleaf project (specifically the Discussion and Limitations section).

Table 1: Summary of interactions with Gemini